Multizetas, after Francis Brown

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0. Introduction

Let $k \geq 0$ and let $s = (s_1, \ldots, s_k)$ be a sequence of k integers ≥ 1. The multizeta number $\zeta(s)$ is the iterated sum (the name is explained in 1.1)

$$\zeta(s) := \sum_{n_1 > \cdots > n_k > 0} \frac{1}{n_1^{s_1} \cdots n_k^{s_k}} \tag{0.1}$$

(the sum being over strictly decreasing sequences of k integers > 0). For $k = 0$, this definition gives $\zeta(s) = 1$. For $k \geq 1$, the sum (0.1) converges if and only if $s_1 \geq 2$. We shall consider only convergent $\zeta(s)$, i.e. those for which (0.1) converges.

The product of two multizeta numbers is a linear combination, with integer coefficients, of multizeta numbers. See 3.6. It is therefore equivalent to determine the polynomial relations with rational coefficients among the multizeta numbers, or to determine the $\mathbb{Q}$-linear relations among them.

For $k = 1$, the convergent $\zeta(s)$ are the values at the integers $n \geq 2$ of Riemann's zeta function. Euler showed that for n even, $\zeta(n)$ is a rational multiple of π^n ([3]). Computing $\zeta(3)$ to 10 significant figures, he also checked that $\zeta(3)$ was not the product of π^3 by a rational number of small denominator. Following a correspondence with Goldbach, Euler studied in [4], for $k = 2$, the variant of the sums (0.1) in which one sums over $n_1 \geq n_2$:

$$\zeta_{\text{Euler}}(s_1, s_2) = \zeta(s_1, s_2) + \zeta(s_1 + s_2).$$

He proves a series of $\mathbb{Q}$-linear relations among multizeta numbers, among them $\zeta(2, 1) = \zeta(3)$. One of his motivations was the hope of obtaining information on the values of ζ at the odd integers ≥ 3 (see the end of his letter to Goldbach of January 5, 1743).

The problem of determining all $\mathbb{Q}$-linear relations among multizeta numbers has two aspects:

(A) What relations should one expect? Prove them.

(B) Prove that there are no others.

The available evidence, both numerical and theoretical, suggests that every $\mathbb{Q}$-linear relation among the $\zeta(s)$ is a sum of isobaric relations, i.e. relations among $\zeta(s)$ of the same weight $\sum s_i$. If one restricts oneself to considering only isobaric relations, one knows nothing about problem (B). We shall speak only of (A).

F. Brown has defined a class of $\mathbb{Q}$-linear relations among the $\zeta(s)$, the motivic relations. More precisely, he defines a graded $\mathbb{Q}$-algebra M, homogeneous elements $\zeta^m(s)$ of M, of degree the weight $\sum s_i$ of s, and a homomorphism $\text{real} : M \to \mathbb{R}$ such that $\text{real}(\zeta^m(s)) = \zeta(s)$. As a vector space, the algebra M is generated by the $\zeta^m(s)$. The motivic relations among the $\zeta(s)$ are those that come from relations among the $\zeta^m(s)$.

A variant of Grothendieck's period conjecture implies that every $\mathbb{Q}$-linear relation among the $\zeta(s)$ is motivic. This prediction has been verified for many of the known relations. Warning: although some proofs of relations among the $\zeta(s)$ are easy to lift to proofs of the same relations among the $\zeta^m(s)$, this is far from always being the case.

Theorem 0.2 (F. Brown [1]). The $\zeta^m(s_1, \ldots, s_k)$ for which each s_i is equal to 2 or 3 form a basis of M over $\mathbb{Q}$.

Each $\zeta(s)$ can therefore be expressed uniquely, in a motivic way, as a linear combination with rational coefficients of $\zeta(s_1, \ldots, s_k)$ with $k \geq 0$ and each $s_i \in \{2, 3\}$. Unfortunately, Brown's proof does not provide an algorithm, or at least not a usable algorithm, for finding what this linear combination is. From Theorem 0.2, F. Brown deduces that the category of mixed Tate motives over $\mathbb{Z}$ is generated by the fundamental group of $\mathbb{P}^1 - \{0, 1, \infty\}$. See 7.18.

Some of our notations differ from those of [1]. The correspondence between the notations is involutive: $\zeta(s_1, \ldots, s_k)$ becomes $\zeta(s_k, \ldots, s_1)$, the composition of paths $\alpha\beta$ becomes $\beta\alpha$, a monomial in the e_t (respectively a sequence of 0 and 1) is to be replaced by the same one read from right to left, and e_1 is to be replaced by $-e_1$.

1. Integral formula

1.1

Let $\omega_1, \dots, \omega_N$ be 1-forms on the interval $[0,1]$. The iterated integral $\mathrm{It}\int_0^1 \omega_1, \dots, \omega_N$ is the integral, over the simplex

$$\sigma_N := \{(t_1, \dots, t_N) \mid 1 \geq t_1 \geq \dots \geq t_N \geq 0\}, \tag{1.1.1}$$

contained in the cube $[0,1]^N$, of

$$\mathrm{pr}_1^* \omega_1 \wedge \mathrm{pr}_2^* \omega_2 \wedge \dots \wedge \mathrm{pr}_N^* \omega_N.$$

Let S be the operation that attaches to a 1-form ω the function

$$S[\omega](t) := \int_0^t \omega.$$

The iterated integral $\mathrm{It}\int_0^1 \omega_1, \dots, \omega_N$ can be constructed as follows: apply S to ω_N, multiply the function obtained by ω_{N-1}, apply S, multiply by ω_{N-2}, and so on; at the end, evaluate at 1:

$$\mathrm{It}\int_0^1 \omega_1, \dots, \omega_N = S[\omega_1 . S[\omega_2 . \cdots S[\omega_N] \cdots]](1). \tag{1.1.2}$$

It is this construction that explains the name "iterated integral". A sum such as (0.1) admits an analogous description in terms of the operator P which, to a function $f(n)$ $(n \geq 1)$, attaches the function

$$P[f](n) := \sum_{m=1}^{n-1} f(m);$$

this explains the name "iterated sum" given to (0.1).

Let us compute in this way the iterated integral

$$\mathrm{It}\int_0^1 \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_1-1)\text{ times}}, \frac{dt}{1-t}, \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_2-1)\text{ times}}, \frac{dt}{1-t}, \dots, \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_k-1)\text{ times}}, \frac{dt}{1-t}. \tag{1.1.3}$$

We have

$$\frac{dt}{1-t} = \sum_{n \geq 0} t^n dt.$$

Applying S term by term, one obtains $\sum_{n \geq 1} t^n/n$. Multiplying by dt/t and applying S, one obtains $\sum_{n \geq 1} t^n/n^2$. Iterating $(s_k - 1)$ times, one obtains $\sum_{n \geq 1} t^n/n^{s_k}$. Multiplying by $dt/(1-t) = \sum_{m \geq 0} t^m dt$ and applying S, one obtains

$$S\left(\sum_{m \geq 0, n \geq 1} \frac{t^{m+n}}{n^{s_k}} dt\right) = \sum_{m,n \geq 1} \frac{t^{m+n}}{(m+n)n^{s_k}} = \sum_{m > n > 0} \frac{t^m}{m n^{s_k}}.$$

Continuing in this way, one finally obtains:

Proposition 1.2. The iterated integral (1.1.3) is equal to $\zeta(s_1, \dots, s_k)$.

1.3

Whereas the notion of an infinite sum is foreign to algebraic geometry, the study of integrals of algebraic quantities is one of its sources. It is thanks to Proposition 1.2 that algebraic geometry, more precisely the theory of mixed Tate motives, is useful in the study of multizeta numbers.

2. Pro-unipotent fundamental group and cohomology

2.1

We shall need a variant Z' of the descending central series Z of a group Γ. For each $i \geq 1$, Γ/Z_i is a nilpotent group. The set T_i of its torsion elements is therefore a subgroup. We define Z'_i as the inverse image of T_i in Γ. Below, we shall always assume Γ to be finitely generated. This hypothesis implies that the Z'_i/Z'_{i+1} are finitely generated free abelian groups.

First suppose Γ nilpotent and torsion-free, i.e. that there is an A such that $Z'_A = 0$. For each i ($1 \leq i < A$), choose a basis of Z'_i/Z'_{i+1} and lift it to Z'_i. Let $\gamma_1, \ldots, \gamma_M$ be the elements of Γ obtained, arranged in an arbitrary order, and let $w(j)$ be the integer i such that γ_j lifts an element of a basis of Z'_i/Z'_{i+1}. One checks by induction on A that the map

$$\mathbb{Z}^M \longrightarrow \Gamma : n \longmapsto \gamma_1^{n_1} \cdots \gamma_M^{n_M}$$

is bijective. In such a "system of coordinates", the group law of Γ is given by polynomials with rational coefficients and integer values on the integers. More precisely, if the coordinate of index j is viewed as having weight $w(j)$, the j-th coordinate of $m \circ n$ is a polynomial of weight $\leq w(j)$ in the m_k and n_l.

The same polynomials define a unipotent algebraic group over $\mathbb{Q}$, the unipotent envelope Γ^{un} of Γ. The group $\Gamma^{\mathrm{un}}(\mathbb{Q})$ of its rational points is $\mathbb{Q}^N$, endowed with the group law given by the same formulae as that of Γ. The group $\Gamma^{\mathrm{un}}(\mathbb{Q})$ is the Malcev completion $\Gamma \otimes \mathbb{Q}$ of Γ.

Let $\mathbb{Q}[\Gamma]$ be the group algebra of Γ, and let I be its augmentation ideal. Quillen gave an elegant description of the affine algebra $O(\Gamma^{\mathrm{un}})$ of Γ^{un}: it is the inductive limit of the duals of the $\mathbb{Q}[\Gamma]/I^N$. More precisely, the inclusion of Γ in $\mathbb{Q}[\Gamma]$ identifies the dual of $\mathbb{Q}[\Gamma]$ with the space of functions $\Gamma \to \mathbb{Q}$, and under this identification the dual of $\mathbb{Q}[\Gamma]/I^{N+1}$ becomes the space of polynomials in the n_j of weight $\leq N$.

For an arbitrary finitely generated Γ, the preceding applies to the Γ/Z'_i, and one defines Γ^{un} as the projective limit group scheme of the $(\Gamma/Z'_i)^{\mathrm{un}}$. The affine algebra of Γ^{un} is still the inductive limit of the duals of the $\mathbb{Q}[\Gamma]/I^N$.

2.2

Let E be a reasonable topological space, for example a complex algebraic variety. Assume E connected. A path γ from a to b, $[0,1] \to E$, gives

$$\gamma_N : [0,1]^N \to E^N$$

and, by restriction to the simplex σ_N of (1.1.1), a singular chain of E^N which is a cycle modulo the union of $b = x_1$, $x_1 = x_2$, …, $x_N = a$. The homology class of this cycle depends only on the homotopy class of γ.

Make $a = b$. Elaborating the preceding construction (cf. [2] §3), one obtains an isomorphism of $\mathbb{Q}[\pi_1(E,a)]/I^{N+1}$ with a relative homology group of E^N and, for its dual $(\mathbb{Q}[\pi_1(E,a)]/I^{N+1})^\vee$, with a relative cohomology group of E^N. From this, by passage to the inductive limit, one obtains a cohomological description of the affine algebra of $\pi_1(E,a)^{\mathrm{un}}$.

No longer suppose that $a = b$. Let $\pi_1(E;b,a)$ be the set of homotopy classes of paths from a to b. The group $\pi_1(E,a)$ acts on the right, by composition of paths, on $\pi_1(E;b,a)$. This action makes $\pi_1(E;b,a)$ a principal homogeneous space, we prefer to say torsor, under $\pi_1(E,a)$. We write $\pi_1(E;b,a)^{\mathrm{un}}$ for the $\pi_1(E,a)^{\mathrm{un}}$-torsor deduced from it by pushing forward by $\pi_1(E,a) \to \pi_1(E,a)^{\mathrm{un}}$. Let $\mathbb{Q}[\pi_1(E;b,a)]$ be the space of formal linear combinations of elements of $\pi_1(E;b,a)$. It is a right $\mathbb{Q}[\pi_1(E,a)]$-module, free of rank one. Set

$$\mathbb{Q}[\pi_1(E;b,a)]/I^N := \mathbb{Q}[\pi_1(E;b,a)]/\mathbb{Q}[\pi_1(E;b,a)]I^N.$$

The preceding construction again provides a cohomological description of the $(\mathbb{Q}[\pi_1(E;b,a)]/I^N)^\vee$, and of the affine algebra of $\pi_1(E;b,a)^{\mathrm{un}}$ which is its inductive limit.

2.3

Let X be an algebraic variety defined over a subfield K of $\mathbb{C}$. The Betti cohomology $H^*_B(X)$ of X is the rational cohomology of the space $X(\mathbb{C})$ of complex points of X, endowed with its classical topology. It is endowed with structures reflecting the algebraic structure of X, in particular with a mixed Hodge structure. One also has the algebraic de Rham cohomology $H^*_{\mathrm{dR}}(X)$. If X is nonsingular, $H^*_{\mathrm{dR}}(X)$ is the hypercohomology of the de Rham complex of sheaves of algebraic differential forms. It is a K-vector space, one has a comparison isomorphism

$$\mathrm{comp}_{B,\mathrm{dR}} : H^i_{\mathrm{dR}}(X) \otimes_K \mathbb{C} \xrightarrow{\sim} H^i_B(X) \otimes_{\mathbb{Q}} \mathbb{C}, \tag{2.3.1}$$

and the filtrations W and F of the mixed Hodge structure of $H^i_B(X)$ come from filtrations of $H^i_{\mathrm{dR}}(X)$. The coefficients of the matrix of the isomorphism (2.3.1), in a basis of $H^i_{\mathrm{dR}}(X)$ and one of $H^i_B(X)$, are called periods.

Thanks to 2.2, all this applies to the affine algebras of the group schemes $\pi_1(X,a)^{\mathrm{un}}$ or of their homogeneous spaces $\pi_1(X;b,a)^{\mathrm{un}}$. We shall write $\pi_1(X,a)_B$ for $\pi_1(X,a)^{\mathrm{un}}$, $\pi_1(X,a)_{\mathrm{dR}}$ for its de Rham variant, and similarly for the $\pi_1(X;b,a)$. We shall put B/dR in the subscript when the statement is valid equally well in Betti and in de Rham.

2.4. Example

Take $K = \mathbb{Q}$ and let X be the multiplicative group $\mathbb{G}_m$, whose group of complex points is $\mathbb{C}^*$. Here the fundamental group, which is commutative, is independent of the base point and is identified with the homology group H_1. In de Rham, the dual H^1_{dR} of H^{dR}_1 is $\mathbb{Q}\,dz/z$. In Betti, $H_1(\mathbb{C}^*, \mathbb{Z})$ is $\mathbb{Z}$, generated by a positive turn around 0. If one does not want to speak of a "positive turn", i.e. to choose which of i and $-i$ is i, it is better to say that $H_1 = \pi_1 = 2\pi i\mathbb{Z}$, with $\ell \in 2\pi i\mathbb{Z}$ corresponding to the loop $\exp(\ell t)$ $(0 \le t \le 1)$, on which the integral of dz/z is ℓ.

The $H^B_1(\mathbb{G}_m) = H_1(\mathbb{C}^*, \mathbb{Q})$ and $H^{\mathrm{dR}}_1(\mathbb{G}_m)$ are the Betti and de Rham realizations of the Tate motive $\mathbb{Q}(1)$ (see §5). We have $\mathbb{Q}(1)_{\mathrm{dR}} = \mathbb{Q}$ (1 being dual to $dz/z \in H^1_{\mathrm{dR}}(\mathbb{G}_m)$), and $\mathbb{Q}(1)_B$, included in $\mathbb{Q}(1)_{\mathrm{dR}} \otimes \mathbb{C}$ by $\mathrm{comp}_{\mathrm{dR},B}$, is $2\pi i\mathbb{Q}$; $\mathbb{Q}(1)$ is purely of Hodge type $(-1,-1)$.

The algebraic group $\pi_1(\mathbb{G}_m, 1)_{\mathrm{dR}}$ is the additive group $\mathbb{G}_a$, whose group of rational points is $\mathbb{Q}(1)_{\mathrm{dR}} = \mathbb{Q}$, and whose affine algebra is $\mathbb{Q}[T]$ (more precisely: $\mathrm{Sym}^*(H^1_{\mathrm{dR}}(\mathbb{G}_m))$). The algebraic group $\pi_1(\mathbb{G}_m, 1)_B$ is the additive group whose group of rational points is $2\pi i\mathbb{Q}$. Affine algebra: $\mathrm{Sym}^*(H^1_B(\mathbb{G}_m))$. It is the algebra of functions f on $2\pi i\mathbb{Z}$, with rational values, and such that, if one identifies $2\pi i\mathbb{Z}$ with $\mathbb{Z}$ by $n \mapsto 2\pi i n$, the following equivalent conditions are verified for N large enough: (a) f is a polynomial of degree $< N$; (b) denoting by Δ the operator $g \mapsto (g(n+1) - g(n))$, one has $\Delta^N f = 0$; (c) the linear form on the group algebra $\mathbb{Q}[\mathbb{Z}]$ defined by f is zero on I^N.

By abuse of language, we shall sometimes write $\mathbb{Q}(1)_{B/\mathrm{dR}}$ for these algebraic groups.

3. The case of $\mathbb{P}^1 - \{0, 1, \infty\}$

3.1

If X is the complement in $\mathbb{P}^1$ of a finite set T of rational points, the variety X is defined over $\mathbb{Q}$ and the mixed Hodge structures of 2.2, 2.3 are Hodge-Tate: Gr^W_n is zero for n odd, and for $n = 2m$ is purely of Hodge type (m,m). For each of the groups M considered in 2.2, let M_{dR} be its de Rham variant (2.3). After renumbering, the weight and Hodge filtrations W and F of M_{dR} are opposite. They define the grading of M_{dR} for which

$$W_{-2n}(M_{\mathrm{dR}}) = \bigoplus_{m \ge n} M^m_{\mathrm{dR}}, \tag{3.1.1}$$
$$F^{-p}(M_{\mathrm{dR}}) = \bigoplus_{m \le p} M^m_{\mathrm{dR}}.$$

The convention (3.1.1) ensures that $\mathbb{Q}(1)_{\mathrm{dR}}$ is of degree one. It is sometimes convenient to grade by the opposite of the degree, the codegree, and to set $(M_{\mathrm{dR}})_m := (M_{\mathrm{dR}})^{-m}$.

Special case: the affine algebra of $\pi_1(X;b,a)_{\mathrm{dR}}$ is graded. It is in degrees ≤ 0 and is reduced to $\mathbb{Q}$ in degree 0. The augmentation morphism $O(\pi_1(X;b,a)_{\mathrm{dR}}) \to \mathbb{Q}$ is a point ${}_b1_a$ of $\pi_1(X;b,a)_{\mathrm{dR}}$. In de Rham, one thus has a canonical path ${}_b1_a$ from one base point to another: for a certain pro-unipotent group scheme Π, each $\pi_1(X;b,a)_{\mathrm{dR}}$ is identified with a copy ${}_b\Pi_a$ of Π, and composition of paths is the group law.

The group Π has the following description. Let L be the Lie algebra over $\mathbb{Q}$ generated by elements e_t $(t \in T)$ subject only to the relation $\sum e_t = 0$. It is endowed with the grading for which each e_t is homogeneous of degree 1. The descending central series Z is the filtration by "degrees $\ge i$". Let $L^\wedge$ be the completion of L for this filtration, and let $\exp(L/Z_i)$ be the unipotent algebraic group with Lie algebra L/Z_i. We have

$$\Pi = \exp(L^\wedge) := \varprojlim \exp(L/Z_i). \tag{3.1.2}$$

Another description: the enveloping algebra A of L is the associative algebra generated by elements e_t $(t \in T)$ subject only to the relation $\sum e_t = 0$. It is graded, each e_t being homogeneous of degree one, and the N-th power of the augmentation ideal I is the part of A of degree $\geq N$. The completion $A^\wedge = \varprojlim A/I^N$ of A is endowed with a coproduct

$$\Delta : A^\wedge \to A^\wedge \widehat{\otimes} A^\wedge : e_t \longmapsto e_t \otimes 1 + 1 \otimes e_t,$$

$L^\wedge$ is the Lie algebra of primitive elements, and for every $\mathbb{Q}$-algebra R, if one sets

$$A^\wedge \widehat{\otimes} R := \varprojlim_N (A/I^N) \otimes_{\mathbb{Q}} R,$$

the group $\Pi(R)$ of R-points of Π is the group of group-like elements of $(A^\wedge \widehat{\otimes} R)^*$:

$$\Pi(R) = \{g \in (A^\wedge \widehat{\otimes} R)^* \mid \Delta g = g \otimes g\}. \tag{3.1.3}$$

The affine algebra $O(\Pi)$ of Π (with the discrete topology) and the completed enveloping algebra $A^\wedge$ (with its projective-limit topology) are dual to one another. The topological dual of $A^\wedge$ is identified with the graded dual $\bigoplus (A_n)^\vee$ of A, and this gives the grading (in degrees ≤ 0) of $O(\Pi)$.

We shall write ω for the logarithmic 1-form on X with values in L:

$$\omega := \sum_{t \neq \infty} e_t \frac{dz}{z-t}. \tag{3.1.4}$$

At each $t \in T$, it has residue e_t.

Suppose that $\infty \in T$ and put $T' := T - \{\infty\}$. Then $A = \mathbb{Q}\langle (e_t)_{t \in T'} \rangle$ and

$$A^\wedge = \mathbb{Q}\langle\langle (e_t)_{t \in T'} \rangle\rangle$$

(associative formal series in the e_t). For every word $w = t_1 \cdots t_k$ in the alphabet T', let $c(w)$ be the linear form on $A^\wedge$ "coefficient of the monomial $e_{t_1} \cdots e_{t_k}$". We again write $c(w)$ for the restriction of $c(w)$ to $\Pi \subset A^{\wedge *}$, and for a formal linear combination of words we put $c(\sum \lambda_w w) = \sum \lambda_w c(w)$. The $c(w)$ form a vector-space basis of $O(\Pi)$. The product in $O(\Pi)$ corresponds to the shuffle product of words:

$$c(w')c(w'') = c(w' \amalg w''), \tag{3.1.5}$$

where, for $w' = t_1 \cdots t_k$ and $w'' = t_{k+1} \cdots t_{k+\ell}$, $w' \amalg w''$ is the sum of the $t_{\sigma^{-1}(1)} \cdots t_{\sigma^{-1}(k+\ell)}$ for σ running through the set $S_{k,\ell}$ of permutations of $\{1, \ldots, k+\ell\}$ increasing on $\{1, \ldots, k\}$ and on $\{k+1, \ldots, k+\ell\}$.

3.2

A path γ from a to b defines a rational point of $\pi_1(X; b, a)_B = \pi_1(X(\mathbb{C}); b, a)^{\mathrm{un}}$. Let $\mathrm{comp}_{\mathrm{dR},B}(\gamma)$ be the image of this rational point by the comparison isomorphism

$$\mathrm{comp}_{\mathrm{dR},B} : \pi_1(X; b, a)_B(\mathbb{C}) \xrightarrow{\sim} \pi_1(X; b, a)_{\mathrm{dR}}(\mathbb{C}) = \Pi(\mathbb{C}).$$

This image is obtained by "integration" along γ, in the noncommutative group Π, of the form ω: it is the value at 1 of the solution $g : [0,1] \to \Pi(\mathbb{C})$, equal to 1 at 0, of

$$dg(t).g(t)^{-1} = \omega \left(\frac{d\gamma}{dt} \right) dt. \tag{3.2.1}$$

3.3

The same formalism applies when one allows the base points a, b to be tangential base points (a nonzero tangent vector at a "rational point at infinity" $t \in T$ of X). Let Θ_t be the tangent space to $\mathbb{P}^1$ at t, $\Theta_t^* := \Theta_t - \{0\}$, and $a \in \Theta_t^*$ a tangential base point at t. Everything happens as if one had a morphism from Θ_t^* to X: one has morphisms

$$\pi_1(\Theta_t^*, a)_{B/\mathrm{dR}} = \mathbb{Q}(1)_{B/\mathrm{dR}} \longrightarrow \pi_1(X, a)_{B/\mathrm{dR}} \tag{3.3.1}$$

compatible with the comparison isomorphisms. On the Betti side, this morphism is induced by the germs of maps φ from $(\Theta_t, 0)$ to $(\mathbb{P}^1, t)$ with differential the identity. The space of these germs is contractible, and each φ induces a map from a small punctured disk around 0 (homotopic to Θ_t^*) to X. In de Rham, and after passage to Lie algebras, (3.3.1) is the morphism from $\mathbb{Q}(1)_{\mathrm{dR}} = \mathbb{Q}$ to $L^\wedge$ which sends 1 to e_t.

The $\mathbb{Q}$-form $\pi_1(X; b, a)_B$ of $\Pi(\mathbb{C})$ is characterized by the property that the images of paths from a to b are rational points.

Warning: $\pi_1(X; b, a)_{\mathrm{dR}}$ is independent of a and b, and $\pi_1(X; b, a)_B$ is locally constant in a and b, but the comparison isomorphism between their complexifications is not locally constant. The independence of $\pi_1(X; b, a)_{\mathrm{dR}}$ from a and b comes from the fact that one has the canonical de Rham path ${}_b1_a$ between any two base points. The image of this path by $\mathrm{comp}_{B,\mathrm{dR}}$ is not locally constant.

3.4

The case that interests us is that in which $X = \mathbb{P}^1 - \{0, 1, \infty\}$ and in which, as base points, one takes the tangential base points 1 at 0 and -1 at 1. They will be denoted 0 and 1. Here

$$\omega = e_0 \frac{dz}{z} + e_1 \frac{dz}{z-1}$$

and for every $\mathbb{Q}$-algebra R,

$$\Pi(R) = \{\text{group-like elements of } R\langle\langle e_0, e_1 \rangle\rangle^*\}$$

(3.1.3). In Betti, one has the "straight path" dch from 0 to 1. We shall again write dch for its image by $\mathrm{comp}_{\mathrm{dR},B}$ in $\Pi(\mathbb{C})$. This image is the following limit for $\epsilon, \eta \to 0$ through values > 0:

$$\mathrm{comp}_{\mathrm{dR},B}(dch) = \lim \exp(-\log \eta\, e_1)\, \mathrm{comp}_{\mathrm{dR},B}([\epsilon, 1-\eta]) \exp(\log \epsilon\, e_0). \tag{3.4.1}$$

The third (respectively first) factor is to be viewed as computing, in the punctured tangent space at 0 (respectively at 1), $\mathrm{comp}_{\mathrm{dR},B}([1, \epsilon])$ (respectively $\mathrm{comp}_{\mathrm{dR},B}([-\eta, -1])$), the form ω being replaced by $(du/u)e_0$ (respectively $(du/u)e_1$).

In general one can express the solution of (3.2.1) as a sum of iterated integrals. Applying 1.2, one obtains that, for convergent $\zeta(s_1, \ldots, s_k)$, one has:

Proposition 3.5. The coefficient of $e_0^{s_1-1} e_1 \cdots e_0^{s_k-1} e_1$ in $dch \in \Pi(\mathbb{C}) \subset \mathbb{C}\langle\langle e_0, e_1 \rangle\rangle^*$ is $(-1)^k \zeta(s_1, \ldots, s_k)$.

This proposition determines the coefficients $c(w)$ of monomials of dch only for w a word in the alphabet $\{0, 1\}$ which does not begin with 1 and does not end with 0. The coefficients $c(0)$ of e_0 and $c(1)$ of e_1 in dch are zero. This, together with the fact that dch is group-like, determines the missing coefficients. More precisely, the vanishing of $c(0)$ and $c(1)$ means that dch belongs to the derived group Π' of Π, whose graded Lie algebra is the part of degree ≥ 2 of that of Π. On Π', writing $\{p\}$ for a power in the monoid of words,

$$c(0^{\{p\}}) = c(1^{\{p\}}) = 0 \quad \text{for } p > 0, \tag{3.5.1}$$

and, applying (3.1.5) to $w10^{\{p-1\}}$ and 0 (respectively to 1 and $1^{\{p-1\}}0w$), one checks by induction on $p \geq 0$ that

$$c(w10^{\{p\}}) = (-1)^p c((w \amalg 0^{\{p\}})1), \tag{3.5.2}$$

respectively

$$c(1^{\{p\}}0w) = (-1)^p c(0(1^{\{p\}} \amalg w))$$

on Π'.

3.6

The identity (3.1.5), evaluated at dch, expresses the product of two multizeta numbers as a linear combination with integer coefficients of multizeta numbers.

4. Motivic multizetas

4.1

Continue to put $X := \mathbb{P}^1 - \{0, 1, \infty\}$ and to take as base points the tangential base points 0 or 1 (see 3.4). In 5.6 we shall give a precise meaning to the expression "motivic structure on the affine algebras of the $\pi_1(X; b, a)_{B/\mathrm{dR}}$". We shall also say "motivic structure on the $\pi_1(X; b, a)_{B/\mathrm{dR}}$". Heuristically, these are structures coming from geometry, and hence having a meaning both for Betti and for de Rham, and respected by the comparison isomorphisms. Examples: the product giving the algebra structure, the coproducts coming from composition of paths, the weight filtration, the isomorphism $\pi_1(X; b, a)_{B/\mathrm{dR}} \to \pi_1(X; \sigma(b), \sigma(a))_{B/\mathrm{dR}}$ induced by the involution $\sigma : x \mapsto 1 - x$ of $\mathbb{P}^1$. Likewise, the morphisms (3.3.1) are a motivic structure on $\mathbb{Q}(1)_{B/\mathrm{dR}}$ and $\pi_1(X, a)_{B/\mathrm{dR}}$. By contrast, neither the Hodge filtration of the affine algebra of $\pi_1(X; b, a)_{\mathrm{dR}}$, nor therefore its grading, nor the independence of $\pi_1(X; b, a)_{\mathrm{dR}}$ from a and b, is motivic. They are available only for de Rham. The structures in question will be structures of multilinear algebra. Let H_B (respectively H_{dR}) be the group scheme over $\mathbb{Q}$ of automorphisms of $\pi_1(X; 1, 0)_B$ (respectively $\pi_1(X; 1, 0)_{\mathrm{dR}}$) respecting all its motivic structures. The weight filtration W of the affine algebra is a filtration by finite-dimensional subspaces, and H_B (respectively H_{dR}) is a projective limit of algebraic subgroups of the $\mathrm{GL}(W_i)$.

4.2

Let $P_{\mathrm{dR},B}$ be the scheme of isomorphisms from $\pi_1(X; 1, 0)_B$ to $\pi_1(X; 1, 0)_{\mathrm{dR}}$ respecting all motivic structures. The group H_B acts on the right on $P_{\mathrm{dR},B}$. The comparison isomorphism $\mathrm{comp}_{\mathrm{dR},B}$ is a complex point of $P_{\mathrm{dR},B}$. The scheme $P_{\mathrm{dR},B}$ is therefore nonempty. It is an H_B-torsor (terminology of 2.2).

If Z_B is a subscheme of $\pi_1(X; 1, 0)_B$ stable under H_B, twisting by the H_B-torsor $P_{\mathrm{dR},B}$ transforms $\pi_1(X; 1, 0)_B$ into $\pi_1(X; 1, 0)_{\mathrm{dR}} = {}_1\Pi_0$, and Z_B into a subscheme Z_{dR} of ${}_1\Pi_0$. Over $\mathbb{C}$, $Z_{\mathrm{dR}}(\mathbb{C})$ is the image of $Z_B(\mathbb{C})$ by $\mathrm{comp}_{\mathrm{dR},B}$.

I like to think of the H_B-orbit of dch as the space of points ch of $\pi_1(X; 1, 0)_B$ having all the properties of dch expressible in motivic terms. For example: the morphism $\pi_1(X; 1, 0)_B \to \pi_1(X; 0, 1)_B$ induced by the involution $x \mapsto 1 - x$ of $\mathbb{P}^1$ must transform ch into its inverse.

Let $M_B \subset \pi_1(X; 1, 0)_B$ be the closure of the H_B-orbit of dch. The twist of M_B by $P_{\mathrm{dR},B}$ is a closed subscheme M_{dR} of ${}_1\Pi_0$. We again write dch for the image of dch by $\mathrm{comp}_{\mathrm{dR},B}$. It is a complex point of M_{dR}, and $M_{\mathrm{dR}}(\mathbb{C}) \subset {}_1\Pi_0(\mathbb{C})$ is the closure of its $H_{\mathrm{dR}}(\mathbb{C})$-orbit.

Definition 4.3. Let $s = (s_1, \dots, s_k)$. If $k \neq 0$, we assume $s_1 \geq 2$. Let w be the word $0^{\{s_1-1\}}1 \cdots 0^{\{s_k-1\}}1$ in the alphabet $\{0, 1\}$, and let $c(w) \in O(\Pi)$ be the function "coefficient of the monomial $e_0^{s_1-1}e_1 \cdots e_0^{s_k-1}e_1$" (3.1). We denote by $\zeta^m(s)$ the restriction to M_{dR} of $(-1)^k c(w)$.

The $\zeta^m(s)$ are the motivic multizetas (F. Brown [1] 2.3). The algebra denoted M in the introduction is $O(M_{\mathrm{dR}})$. Before [1], only the restriction of the $\zeta^m(s)$ to the orbit of 1 was considered. This gave rise to painful contortions. By 3.5, the value of $\zeta^m(s)$ at the complex point dch of M_{dR} is $\zeta(s)$. Every polynomial identity with rational coefficients in the $\zeta^m(s)$ therefore gives, by evaluation at dch, a polynomial relation among the $\zeta(s)$. Grothendieck's yoga of periods predicts that every polynomial relation among the $\zeta(s)$ comes from geometry; more precisely, that it is obtained as above.

For every word w in 0 and 1, similarly write $c^m(w)$ for the restriction of $c(w)$ to M_{dR}. Relying on the remark following 3.5, one can show that the $c^m(w)$ are linear combinations with integer coefficients of the $\zeta^m(s)$.

By construction, H_{dR} acts on M_{dR}. Its Lie algebra therefore acts by derivations on the affine algebra of M_{dR}, a quotient of that of Π. These derivations, which have a meaning only once one has lifted from the $\zeta(s)$ to the $\zeta^m(s)$, will play a key role in F. Brown's arguments.

5. Mixed Tate motives

5.1

We are here in one of the cases in which the philosophy of motives is not only a precious guide, but allows proofs. Over every field K, Voevodsky has defined a motivic triangulated category V_K, endowed with a tensor product, and containing Tate objects $\mathbb{Z}(n)$ ($n \in \mathbb{Z}$). If K is a number field, after tensoring with $\mathbb{Q}$ one can construct a subcategory $MT(K)$ of $V_K \otimes \mathbb{Q}$, the category of mixed Tate motives over K, and this category is Tannakian. This rests on our knowledge of the K-theory groups of a number field, due to the transcendental determination of $H_i(B\,\mathrm{GL}(n,K),\mathbb{C})$ for $n \gg i$.

All this is for me a black box. It is difficult to open, and this is why the identities promised by 0.2 are not explicit.

5.2

Let us review some properties of the category $MT(K)$. It is a Tannakian category over $\mathbb{Q}$. It contains a rank-one object $\mathbb{Q}(1)$, the Tate motive. Since this motive has rank one, $\mathbb{Q}(n) := \mathbb{Q}(1)^{\otimes n}$ is defined for $n \in \mathbb{Z}$. The objects of $MT(K)$ are endowed with a finite increasing functorial filtration, compatible with the tensor product, the weight filtration W. It is indexed by even integers. The functor $M \mapsto \mathrm{Gr}^W(M)$ is exact, and $\mathrm{Gr}^W_{-2n}(M)$ is a sum of copies of $\mathbb{Q}(n)$.

Suppose, for simplicity, that $K = \mathbb{Q}$. Then one has Betti and de Rham fiber functors

$$\omega_B, \omega_{\mathrm{dR}} : MT(K) \longrightarrow \mathbb{Q}\text{-vector spaces}$$

and a comparison isomorphism

$$\mathrm{comp}_{\mathrm{dR},B} : \omega_B \otimes \mathbb{C} \longrightarrow \omega_{\mathrm{dR}} \otimes \mathbb{C}.$$

Note that, because $K = \mathbb{Q}$, the fiber functor ω_{dR} coincides with the fiber functor ω of [2]. For the motive $\mathbb{Q}(1)$, $\omega_B(\mathbb{Q}(1))$, $\omega_{\mathrm{dR}}(\mathbb{Q}(1))$, and $\mathrm{comp}_{\mathrm{dR},B}$ are as in 2.4.

The functor ω_{dR} is endowed with a Hodge filtration F, opposite, up to a renumbering, to the weight filtration of $\omega_{\mathrm{dR}}(M)$ by the $\omega_{\mathrm{dR}}(W_{2n}M)$. This makes it into a graded functor, as in 3.1.

5.3

If T is a Tannakian category, we shall call a T-algebra an Ind-object A of T, endowed with a product $A \otimes A \to A$. One defines the category of affine T-schemes as the opposite of that of commutative T-algebras with unit, and an affine T-group scheme as a group in this category. For T a category of motives, one replaces the prefix T by the adjective motivic.

5.4

The $\pi_1(X;b,a)_B$, $\pi_1(X;b,a)_{\mathrm{dR}}$ of 3.1, 3.3, and the comparison isomorphism between their complexifications are motivic: they are images of a motivic group scheme $\pi_1(X;b,a)_{\mathrm{mot}}$ ([2] §4). The morphisms "composition of paths" and (3.3.1) are motivic as well.

If $X = \mathbb{P}^1 - \{0,1,\infty\}$ and a, b are chosen among the tangential base points $0, 1$ of 3.4, the points $0, 1, \infty$ remain distinct after reduction modulo p, and the tangent vectors chosen at 0 and 1 remain nonzero. One can deduce that the affine algebras $O(\pi_1(X;b,a)_{\mathrm{mot}})$ are Ind-objects of a Tannakian subcategory $MT(\mathbb{Z})$ of $MT(\mathbb{Q})$, that of mixed Tate motives with good reduction over $\mathrm{Spec}(\mathbb{Z})$ ([2] 1.7, 1.4, 1.8). In $MT(\mathbb{Z})$,

$$\mathrm{Ext}^1(\mathbb{Q}(0),\mathbb{Q}(n)) \text{ is } \begin{cases} \text{one-dimensional over } \mathbb{Q}, & n \text{ odd and } n \geq 3, \\ 0, & \text{otherwise,} \end{cases} \tag{5.4.1}$$

$$\mathrm{Ext}^2(\mathbb{Q}(0),\mathbb{Q}(n)) = 0. \tag{5.4.2}$$

5.5

Let T be a Tannakian category over a field K, and let ω be a fiber functor over an extension K' of K. The family of the $\omega(Y)$, for Y in T, endowed with the $\omega(f) : \otimes\omega(Y_i) \to \otimes\omega(Z_j)$ for $f : \otimes Y_i \to \otimes Z_j$ a morphism of T, is a family of K'-vector spaces endowed with a structure of multilinear algebra. The subgroup scheme of $\prod \mathrm{GL}(\omega(Y))$ (product over $\mathrm{Ob}(T)$) that respects this structure is the group scheme $\mathrm{Aut}(\omega)$ of automorphisms of the fiber functor ω. It is denoted $\pi(T,\omega)$ and is called the fundamental group of T at ω. One will see in examples that it is of the "same nature" as the $\omega(Y)$, for example that its affine algebra is graded if the functor ω is graded. The explanation of this fact is that there exists a T-group scheme $\pi(T)$, acting functorially on each object of T, such that for every fiber functor ω, $\pi(T,\omega)$ is $\omega(\pi(T))$.

Suppose $K' = K$. In this case, ω induces an equivalence of T with the category of finite-dimensional linear representations of $\pi(T,\omega)$. If T is a category of motives, the fundamental groups are also called motivic Galois groups.

5.6

Let $\langle(Y_i)\rangle$ be the Tannakian subcategory of T generated (by subquotients, sums, duals, and tensor products) by a family (Y_i) of objects of T. A T-structure on the family of the $\omega(Y_i)$ is the image by ω of a morphism of $\langle(Y_i)\rangle$. More precisely, it is the datum of the image by ω of subquotients of sums of tensor products of the Y_i and $Y_i^\vee$, and of morphisms between such subquotients. If T is a category of motives, one will rather say motivic structure. The subgroup of $\prod \mathrm{GL}(\omega(Y_i))$ which respects all T-structures is therefore the fundamental group of $\langle(Y_i)\rangle$ at ω. Every algebraic subgroup of a group $\mathrm{GL}(V)$ is the subgroup respecting a few subspaces and elements in tensor spaces on V. The equivalence 5.5 therefore ensures that this fundamental group of $\langle(Y_i)\rangle$ at ω is the quotient of $\pi(T,\omega)$ which acts faithfully on the family of the $\omega(Y_i)$. Same terminology if one starts from a family of ind-objects of T: one replaces it by the family of subobjects in T.

Take $T := MT(\mathbb{Z})$. With the notation B/dR of 2.3, set

$$G_{B/\mathrm{dR}} := \pi(MT(\mathbb{Z}), \omega_{B/\mathrm{dR}}). \tag{5.6.1}$$

The group scheme $H_{B/\mathrm{dR}}$ of automorphisms of $\pi_1(X;1,0)_{B/\mathrm{dR}}$ "respecting all motivic structures" considered in 4.1 is the quotient of $G_{B/\mathrm{dR}}$ acting faithfully on

$$\pi_1(X;1,0)_{B/\mathrm{dR}} = \omega_{B/\mathrm{dR}}(\pi_1(X;1,0)_{\mathrm{mot}}).$$

By the equivalence 5.5, a closed subscheme of $\pi_1(X;1,0)_B$ stable under H_B is the Betti realization of a closed subscheme of $\pi_1(X;1,0)_{\mathrm{mot}}$. Example: there is a unique motivic subscheme M_{mot} of $\pi_1(X;1,0)_{\mathrm{mot}}$ whose Betti realization is the orbit closure M_B; the subscheme M_{dR} (4.2) of $\pi_1(X;1,0)_{\mathrm{dR}}$ is its de Rham realization.

5.7

The action of $G_{B/\mathrm{dR}}$ on $\mathbb{Q}(1)_{B/\mathrm{dR}}$ defines a morphism t from $G_{B/\mathrm{dR}}$ to the multiplicative group $\mathbb{G}_m$. For every M in $MT(\mathbb{Z})$, $G_{B/\mathrm{dR}}$ respects the weight filtration of $\omega_{B/\mathrm{dR}}M$, and since $\mathrm{Gr}^W_{-2n}(M)$ is a sum of copies of $\mathbb{Q}(n)$, the action of g in $G_{B/\mathrm{dR}}$ on $\mathrm{Gr}^W_{-2n}(M)$ is multiplication by $t(g)^n$. The kernel $U_{B/\mathrm{dR}}$ of t therefore acts trivially on the $\mathrm{Gr}^W_{-2n}(M)$: it is a pro-unipotent group scheme.

Let $\tau(\lambda)$ be multiplication by λ^n on $\omega^n_{\mathrm{dR}}(M)$. The $\tau(\lambda)$ define a section $\tau : \mathbb{G}_m \to G_{\mathrm{dR}}$ of t, which makes G_{dR} a semidirect product $\mathbb{G}_m \ltimes U_{\mathrm{dR}}$. The action of $\mathbb{G}_m$ on U_{dR} (by inner automorphisms in G_{dR}) will again be denoted τ. It permits one to define the graded Lie algebra $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$. The vanishing (5.4.2) implies that this graded Lie algebra is free, and (5.4.1) that it is freely generated by one generator in each odd degree ≥ 3. The functor ω_{dR} induces an equivalence of categories from $MT(\mathbb{Z})$ with the category of finite-dimensional graded vector spaces, endowed with an action, compatible with the gradings and therefore nilpotent, of $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$.

5.8

Let P be the scheme of isomorphisms of fiber functors from ω_{dR} to ω_B. The comparison isomorphism $\mathrm{comp}_{B,\mathrm{dR}}$ is a complex point of P. The scheme P is therefore nonempty, and is a G_{dR}-torsor. Since G_{dR} is an extension of $\mathbb{G}_m$ by U_{dR}, a Galois cohomology argument shows that every G_{dR}-torsor has a rational point: there exists an isomorphism of fiber functors $p : \omega_{\mathrm{dR}} \to \omega_B$. Write

$$p = \mathrm{comp}_{B,\mathrm{dR}} \cdot a \tag{5.8.1}$$

with $a \in G_{\mathrm{dR}}(\mathbb{C})$. It follows from (5.8.1) that for every M in $MT(\mathbb{Z})$, if one identifies $\omega_{\mathrm{dR}}(M)\otimes\mathbb{C}$ with $\omega_B(M)\otimes\mathbb{C}$ by $\mathrm{comp}_{B,\mathrm{dR}}$, $\omega_B(M) \subset \omega_B(M)\otimes\mathbb{C} = \omega_{\mathrm{dR}}(M)\otimes\mathbb{C}$ verifies

$$\omega_B(M) = a(\omega_{\mathrm{dR}}(M)). \tag{5.8.2}$$

Since $\omega_B(\mathbb{Q}(1)) = 2\pi i\mathbb{Q}$, one has $t(a) \in 2\pi i\mathbb{Q}^*$, and $a = u\tau(\lambda)$ with $u \in U_{\mathrm{dR}}(\mathbb{C})$ and $\lambda \in 2\pi i\mathbb{Q}^*$. A compatibility with complex conjugation makes it possible to show that one can choose ([2] 2.12)

$$a \in U_{\mathrm{dR}}(\mathbb{R}).\tau(2\pi i). \tag{5.8.3}$$

6. The space $\widetilde{H}(2,3)$ of functions on $\pi_1(X;1,0)_{\mathrm{dR}}$

From now on, $X := \mathbb{P}^1 - \{0,1,\infty\}$, and the notation of 3.4 is in force. By abuse of language, write $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$ (respectively $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle^*$) for the group scheme over $\mathbb{Q}$ such that, for every $\mathbb{Q}$-algebra R, the set of its R-points is $R\langle\langle e_0, e_1\rangle\rangle$ (respectively $R\langle\langle e_0, e_1\rangle\rangle^*$). Recall that $\pi_1(X;b,a)_{\mathrm{dR}}$ is a copy ${}_b\Pi_a$ of the subscheme Π of group-like elements (3.1.3) of $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle^*$. We shall regard it as contained in a copy $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle_{b,a}$ of $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$. A point x of $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$ will sometimes be denoted ${}_bx_a$ when regarded as belonging to this copy.

6.1

As a $\mathbb{Q}$-vector space, the affine algebra $O(\Pi)$ of Π has for basis the "coefficients of monomials" $c(w)$ of 3.1. It is graded. If the word w in the alphabet $\{0,1\}$ has length d, the element $c(w)$ of $O(\Pi)$ is of degree $-d$, of codegree d. When we write B', with decorations attached, for a set of words, B, with the same decorations attached, will be the corresponding set of $c(w)$.

Definition 6.2.

(i) $B'(2,3)$ is the set of words which are concatenations of the words 01 and 001;

(ii) $\widetilde{H}(2,3)$ is the vector subspace of $O(\Pi)$ with basis $B(2,3)$.

The restrictions to M_{dR} of the $c(w)$ in $B(2,3)$ are those of the $(-1)^k\zeta^m(s_1,\dots,s_k)$ where each s_i is equal to 2 or 3.

For any graded vector space Z, with finite-dimensional homogeneous components, the Poincaré series of Z is

$$P(Z,t) = \sum \dim(Z_d)t^d. \tag{6.2.1}$$

If $\widetilde{H}(2,3)$ is graded by codegree, then

$$P(\widetilde{H}(2,3),t) = (1-t^2-t^3)^{-1}. \tag{6.2.2}$$

This is checked by expanding $(1-t^2-t^3)^{-1} = \sum(t^2+t^3)^n$.

6.3

The words $w \in B'(2,3)$ are characterized by:

a. they do not begin with 1 and do not end with 0;

b. they do not contain two consecutive 1's;

c. they do not contain three consecutive 0's.

That a function $f \in O(\Pi)$ is a linear combination of $c(w)$ for w satisfying a, b, or c is respectively equivalent to:

a'. for γ in Π, $f(\exp(ue_1)\gamma\exp(te_0))$ is independent of t and u;

b'. for γ,δ in Π, $f(\gamma\exp(te_1)\delta)$ has degree ≤ 1 in t;

c'. for γ,δ in Π, $f(\gamma\exp(te_0)\delta)$ has degree ≤ 2 in t.

To verify this, note that linear combinations of elements of $\Pi(\mathbb{Q})$ are dense in $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$, and that, if f is interpreted as a continuous linear form on $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$, condition a' (respectively b', c') is equivalent to the same condition where γ,δ are allowed to be arbitrary in $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$. Conditions a', b', c' are motivic, since they can be expressed in terms of composition of paths and of (3.3.1). Here γ (respectively γ,δ) must be interpreted as lying in ${}_1\Pi_0$ (respectively in ${}_1\Pi_1$ and ${}_1\Pi_0$, respectively in ${}_1\Pi_0$ and ${}_0\Pi_0$).

To see the motivic nature of a', b', c', one may observe that these conditions have a Betti formulation. This is obtained by viewing f as a function on $\pi_1(X;1,0)$ and replacing $\exp(te_0)$ (respectively $\exp(te_1)$) by σ_0^t (respectively σ_1^t), where $t \in \mathbb{Z}$ and σ_0 (respectively σ_1) is a positive turn around 0 in the punctured tangent space to $\mathbb{P}^1$ at 0 (respectively at 1) (cf. 3.3).

We leave to the reader the verification of the following lemma.

Lemma 6.4. Let $f = \sum \lambda_w c(w)$ in $O(\Pi)$. In order that, for all $\gamma_0, \ldots, \gamma_r$ in Π,

$$f(\gamma_0 \exp(te_0)\gamma_1 \exp(te_0) \cdots \exp(te_0)\gamma_r) \tag{6.4.1}$$

have degree $\leq d$ in t, it is necessary and sufficient that, for every w such that $\lambda_w \neq 0$, the total length of at most r disjoint strings of consecutive 0's in w be $\leq d$.

6.5

Let $B'_\ell(2,3)$ (respectively $B'_{\leq \ell}(2,3)$) be the set of words w in $B'(2,3)$ having exactly (respectively at most) ℓ factors 001. With the convention of 6.1, let N_ℓ be the subspace of $\widetilde{H}(2,3)$ with basis $B_{\leq \ell}(2,3)$. It follows from 6.4 that, in order for $f \in \widetilde{H}(2,3)$ to lie in N_ℓ, it is necessary and sufficient that, for $r = \ell + 1$ and every choice of the γ_i, (6.4.1) have degree $\leq 2\ell + 1$ in t. This property is motivic. Hence the action of G_{dR} on $O({}_1\Pi_0)$ preserves $\widetilde{H}(2,3)$ and its increasing filtration N.

Proposition 6.6. The action of U_{dR} on $\mathrm{Gr}^N \widetilde{H}(2,3)$ is trivial.

Proof. The graded Lie algebra of U_{dR} is generated by elements ν_d of odd degrees $d \geq 3$, and, if $O({}_1\Pi_0)$ is graded by degree (6.1), the action is compatible with gradings. It is enough to show that each ν_d acts trivially. Let w be a word of length n, a concatenation of words 01 and of ℓ words 001. Then $n \equiv \ell \pmod 2$. The image of $c(w)$ under ν_d is a linear combination of $c(w')$, with w' of length $n - d$ and a concatenation of words 01 and of $\ell' \leq \ell$ words 001. We have $n - d \equiv \ell' \pmod 2$, and, since d is odd, ℓ and ℓ' do not have the same parity. Thus $\ell' < \ell$, proving 6.6.

6.7

It follows from 6.6 that the action of $\mathrm{Lie}^{\mathrm{gr}}\, U_{\mathrm{dR}}$ on $N_\ell / N_{\ell-2}$ factors through the abelianization $\mathrm{Lie}^{\mathrm{gr}}\, U_{\mathrm{dR}}^{\mathrm{ab}}$. The action of ν_d on $N_\ell/N_{\ell-2}$ therefore does not depend, up to a scalar, on the choice of generators ν_d. It is given by a map

$$\mathrm{Gr}^N(\nu_d) : \mathrm{Gr}^N_\ell(\widetilde{H}(2,3)) \longrightarrow \mathrm{Gr}^N_{\ell-1}(\widetilde{H}(2,3)). \tag{6.7.1}$$

The vector space $\mathrm{Gr}^N_\ell(\widetilde{H}(2,3))$ has as basis the image of $B_\ell(2,3)$ (6.5). It is graded by codegree. Let $B'_\ell(2,3)_n$ be the part of $B'_\ell(2,3)$ formed by words of length n. Write $\mathrm{Gr}^N_\ell(\widetilde{H}(2,3))_n$ for the direct factor of codegree n in $\mathrm{Gr}^N_\ell(\widetilde{H}(2,3))$. It has as basis the image of $B_\ell(2,3)_n$. It is non-zero only when $n \equiv \ell \pmod 2$ and $n - 3\ell \geq 0$.

The morphisms (6.7.1) induce

$$\sum_d \mathrm{Gr}^N(\nu_d) : \mathrm{Gr}^N_\ell(\widetilde{H}(2,3))_n \longrightarrow \bigoplus_d \mathrm{Gr}^N_{\ell-1}(\widetilde{H}(2,3))_{n-d}, \tag{6.7.2}$$

where it is enough to sum over odd d with $3 \leq d \leq 3 + n - 3\ell$. The left-hand side of (6.7.2) has a basis indexed by $B'_\ell(2,3)_n$. The right-hand side has a basis indexed by the union of the $B'_{\ell-1}(2,3)_{n-d}$. If $\ell \geq 1$, one obtains a bijection between these sets of words by associating to a word $w \in B'_\ell(2,3)_n$ the word $\bar{w}$ obtained from w by deleting its final $(001)(01)\cdots(01)$.

F. Brown computes the morphisms (6.7.2) and proves the following theorem.

Theorem 6.8. For $\ell \geq 1$, the morphisms (6.7.2) are isomorphisms.

The strategy he uses to prove 6.8 will be explained in §8.

Corollary 6.9. The restriction morphism to M_{dR},

$$\mathrm{restr} : \widetilde{H}(2,3) \longrightarrow O(M_{\mathrm{dR}}), \tag{6.9.1}$$

is injective.

Applying (6.2.2), one deduces from 6.9 the following corollary.

Corollary 6.10. The Poincaré series of $O(M_{\mathrm{dR}})$ satisfies

$$P(O(M_{\mathrm{dR}}), t) \geq (1 - t^2 - t^3)^{-1} \tag{6.10.1}$$

(coefficientwise), with equality if and only if (6.9.1) is bijective.

Assume 6.8, and prove 6.9.

Lemma 6.11. The morphism (6.9.1) is injective on N_0.

Proof. The morphism induced by (6.9.1) from N_0 to $O(M_{\mathrm{dR}})$ is compatible with the codegree gradings. It is therefore enough to verify injectivity in each codegree separately. The elements $c((01)^n)$ form a basis of N_0, and $c((01)^n)$ has codegree $2n$. Since these codegrees are distinct, it is enough to check that, for every n, the restriction of $c((01)^n)$ to M_{dR} is non-zero. The value of $c((01)^n)$ at $dch \in M_{\mathrm{dR}}(\mathbb{C})$ is $(-1)^n \zeta(2\{n\})$, where $\zeta(2\{n\}) = \zeta(2, \ldots, 2)$ with n occurrences of "2". We conclude by observing that every multizeta number is > 0, hence non-zero.

Remark 6.12. Expanding Euler's identities

$$\prod_{n>0} (1 - z^2/n^2) = \sin(\pi z)/(\pi z) = \sum (-1)^n (\pi z)^{2n}/(2n+1)!,$$

one sees that $\zeta(2\{n\}) = \pi^{2n}/(2n+1)!$. That $\zeta(2\{n\})$ is a rational multiple of π^{2n} already followed from the fact that the motivic subspace N_0 of $O({}_1\Pi_0)$ is fixed under U_{dR}, hence is the de Rham realization of a sum of Tate motives. This forces $\zeta(2\{n\})$ to be a period of $\mathbb{Q}(-2n)$.

6.13. Proof that 6.8 implies 6.9

Filter the image $\mathrm{restr}(\widetilde{H}(2,3))$ of $\widetilde{H}(2,3)$ in $O(M_{\mathrm{dR}})$ by the images of the N_ℓ. It is enough to prove the injectivity of

$$\widetilde{H}(2,3) \longrightarrow \mathrm{restr}(\widetilde{H}(2,3)) \tag{6.13.1}$$

after passing to associated gradeds. We prove by induction on $\ell \geq 0$ the injectivity on $\mathrm{Gr}^N_\ell(\widetilde{H}(2,3))$. Lemma 6.11 proves it for $\ell = 0$. If $\ell > 0$, consider the morphism

$$\sum \nu_d : \mathrm{Gr}^N_\ell \widetilde{H}(2,3) \longrightarrow \bigoplus \mathrm{Gr}^N_{\ell-1} \widetilde{H}(2,3),$$

where the target is a sum of copies of $\bigoplus \mathrm{Gr}^N_{\ell-1} \widetilde{H}(2,3)$ indexed by the odd integers $d \geq 3$. Theorem 6.8 says that this morphism is bijective. The morphism (6.9.1) commutes with the derivations ν_d. Thus the diagram

$$\begin{tikzcd}
\mathrm{Gr}^N_\ell \widetilde{H}(2,3) \arrow[r, "\mathrm{Gr}^N_\ell (6.13.1)"] \arrow[d, "\sum \nu_d"', "\sim"] & \mathrm{Gr}^N_\ell \mathrm{restr}(\widetilde{H}(2,3)) \arrow[d, "\sum \nu_d"] \\
\bigoplus \mathrm{Gr}^N_{\ell-1} \widetilde{H}(2,3) \arrow[r, "\mathrm{Gr}^N_{\ell-1}(6.13.1)"'] & \bigoplus \mathrm{Gr}^N_{\ell-1} \mathrm{restr}(\widetilde{H}(2,3))
\end{tikzcd}$$

is commutative. The induction hypothesis says that $\mathrm{Gr}^N_{\ell-1}(6.13.1)$ is injective. Since the first vertical arrow is injective, $\mathrm{Gr}^N_\ell(6.13.1)$ is injective as well.

7. Structure of the orbit closure M_{dR}

7.1

Consider the groupoid of the ${}_b\Pi_a$ for $a, b \in \{0, 1\}$, endowed with $e_0 \in \mathrm{Lie}({}_0\Pi_0)$ and $e_1 \in \mathrm{Lie}({}_1\Pi_1)$. Let V_{dR} be the group scheme of automorphisms of this structure. The action of U_{dR} on the groupoid of the ${}_b\Pi_a$ factors through a morphism

$$I : U_{\mathrm{dR}} \longrightarrow V_{\mathrm{dR}}. \tag{7.1.1}$$

We write IU_{dR} for the image of U_{dR} in V_{dR}.

The group of automorphisms that preserve $e_0 \in \mathrm{Lie}({}_0\Pi_0)$ and $e_1 \in \mathrm{Lie}({}_1\Pi_1)$ only up to a scalar factor, the same factor for e_0 and e_1, is a semidirect product $\mathbb{G}_m \ltimes V_{\mathrm{dR}}$. Its group law is denoted $\circ$, and τ denotes the inclusion of $\mathbb{G}_m$. The action of $\mathbb{G}_m$ on the ${}_b\Pi_a$ is the action of the subgroup $\mathbb{G}_m$ of G_{dR}. It corresponds to the gradings.

By [2] 5.9,

$$v \longmapsto v({}_1 1_0) : V_{\mathrm{dR}} \longrightarrow {}_1\Pi_0 \tag{7.1.2}$$

is an isomorphism of schemes. This isomorphism identifies V_{dR} with Π, endowed with a new group law, denoted $\circ$. The action τ of $\mathbb{G}_m$ defining the grading of $O({}_1\Pi_0)$ fixes ${}_1 1_0$. It follows that (7.1.2) is compatible with gradings. Since it sends the identity to the identity, it defines an isomorphism of graded vector spaces

$$\mathrm{Lie}^{\mathrm{gr}}\, V_{\mathrm{dR}} \xrightarrow{\sim} \mathrm{Lie}^{\mathrm{gr}}\, \Pi. \tag{7.1.3}$$

Thus the graded Lie algebra $\mathrm{Lie}^{\mathrm{gr}}\, V_{\mathrm{dR}}$ is identified with $\mathrm{Lie}^{\mathrm{gr}}\, \Pi$, endowed with a new bracket, the Ihara bracket $\{,\}$.

7.2

Write $x \mapsto \gamma_{b,a}\langle x\rangle$ for the action on ${}_b\Pi_a$ of γ in $(\Pi, \circ)$. It is computed as follows. The action on ${}_a\Pi_a$ preserves the group structure. For $a = 0$, it is induced by the automorphism

$$\gamma_{00} : e_0 \longmapsto e_0, \qquad e_1 \longmapsto \gamma^{-1} e_1 \gamma \tag{7.2.1}$$

of the completed enveloping algebra $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$ of $\mathrm{Lie}\,\Pi$. To check this, note that $e_0 \in \mathrm{Lie}({}_0\Pi_0)$ and $e_1 \in \mathrm{Lie}({}_1\Pi_1)$ are fixed by definition of V_{dR}. Writing ${}_0(e_1)_0 = ({}_1 1_0)^{-1}{}_1(e_1)_1({}_1 1_0)$, one obtains as asserted ${}_0(e_1)_0 \mapsto \gamma^{-1}{}_1(e_1)_1\gamma$.

The automorphism γ_{10} of ${}_1\Pi_0$ is induced by the γ_{00}-semilinear automorphism

$$\gamma_{10} : x \longmapsto \gamma\, \gamma_{00}\langle x\rangle \tag{7.2.2}$$

of the right $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$-module $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$. This is checked as above, starting from ${}_1x_0 = {}_1 1_0\, {}_0x_0$. Since $\gamma \circ x$ sends ${}_1 1_0$ to $\gamma_{10}\langle {}_1x_0\rangle$, it is also

$$\gamma_{10} : x \longmapsto \gamma \circ x,$$

and hence

$$x \circ y = x\, x_{00}\langle y\rangle. \tag{7.2.3}$$

7.3

Via $\mathrm{comp}_{\mathrm{dR},B}$, regard $\pi_1(X; b, a)_B$ as a $\mathbb{Q}$-structure on the complexification of $\pi_1(X; b, a)_{\mathrm{dR}} = {}_b\Pi_a$, the Betti $\mathbb{Q}$-structure, denoted $({}_b\Pi_a)_B$. For this new $\mathbb{Q}$-structure, $2\pi i e_0$ in $\mathrm{Lie}({}_0\Pi_0)_{\mathbb{C}}$ and $2\pi i e_1$ in $\mathrm{Lie}({}_1\Pi_1)_{\mathbb{C}}$ are rational: their exponentials are σ_0 and σ_1 of 6.3. We deduce the following lemma.

Lemma 7.4. Let ch be a rational point of $({}_1\Pi_0)_B$. View its image by $\mathrm{comp}_{\mathrm{dR},B}$, still denoted ch, as a complex point of $(\Pi, \circ)$. Then the complex point $ch \circ \tau(2\pi i)$ of $\mathbb{G}_m \ltimes V_{\mathrm{dR}} = \mathbb{G}_m \ltimes (\Pi, \circ)$ transforms the $\mathbb{Q}$-structure of ${}_b\Pi_a$ into its Betti $\mathbb{Q}$-structure.

7.5

Let $a = u\tau(2\pi i) \in U_{\mathrm{dR}}(\mathbb{R})\tau(2\pi i) \subset G_{\mathrm{dR}}(\mathbb{C})$ as in (5.8.3). If ch is as in 7.4, both $ch\tau(2\pi i)$ and $I(u)\tau(2\pi i)$ transform the $\mathbb{Q}$-structure of the ${}_b\Pi_a$ into their Betti $\mathbb{Q}$-structure. If $\gamma \in V_{\mathrm{dR}}(\mathbb{C}) = \Pi(\mathbb{C})$ is defined by

$$ch \circ \tau(2\pi i) = I(u) \circ \tau(2\pi i) \circ \gamma, \tag{7.5.1}$$

then γ preserves the $\mathbb{Q}$-structure, hence lies in $V_{\mathrm{dR}}(\mathbb{Q}) = \Pi(\mathbb{Q})$. Still writing τ for the action of $\mathbb{G}_m$ on Π, one may rewrite (7.5.1) as

$$ch = I(u) \circ \tau(2\pi i)(\gamma).$$

Apply this to the straight path dch.

Lemma 7.6. For u as above, there exists γ in $\Pi(\mathbb{Q})$ such that

$$dch = I(u) \circ \tau(2\pi i)(\gamma). \tag{7.6.1}$$

Every such γ satisfies

$$\tau(-1)(\gamma) = \gamma. \tag{7.6.2}$$

Proof. Formula (7.6.2) follows from the fact that u and dch, and hence also $\tau(2\pi i)(\gamma)$, are real.

7.7

Apply (7.2.3) to (7.6.1). The orbit of dch under $U_{\mathrm{dR}}(\mathbb{C})$ is

$$IU_{\mathrm{dR}}(\mathbb{C}) \circ \tau(2\pi i)(\gamma) \subset \Pi_{\mathrm{dR}}(\mathbb{C}).$$

To obtain the orbit under $G_{\mathrm{dR}} = \mathbb{G}_m \ltimes U_{\mathrm{dR}}$, it is enough to replace $\tau(2\pi i)(\gamma)$ by its orbit $\tau(\mathbb{C}^*)(\gamma)$ under $\mathbb{G}_m$:

$$G_{\mathrm{dR}}(\mathbb{C})(dch) = IU_{\mathrm{dR}}(\mathbb{C}) \circ \tau(\mathbb{C}^*)(\gamma). \tag{7.7.1}$$

Indeed, IU_{dR} is already stable under $\mathbb{G}_m$. As was already known, the orbit (7.7.1) is, in the de Rham sense, defined over $\mathbb{Q}$. It is the set of complex points of

$$IU_{\mathrm{dR}} \circ \tau(\mathbb{G}_m)(\gamma) \subset \Pi. \tag{7.7.2}$$

In (7.7.2), the product is that of $(\Pi, \circ) = V_{\mathrm{dR}}$, and the result is regarded as a subscheme of ${}_1\Pi_0$.

In $\mathbb{Q}\langle\langle e_0, e_1\rangle\rangle$, write $\gamma = \sum c_w w$. By (7.6.2), the sum runs only over monomials w of even degree. If $\rho(\lambda) := \sum \lambda^{\deg(w)/2} c_w w$, then $\rho(\lambda)$ is defined for $\lambda \in \mathbb{G}_a$ and satisfies

$$\rho(\lambda^2) = \tau(\lambda)(\gamma). \tag{7.7.3}$$

Proposition 7.8. The morphism of schemes

$$IU_{\mathrm{dR}} \times \mathbb{G}_a \longrightarrow \Pi, \qquad (u, \lambda) \longmapsto u \circ \rho(\lambda) \tag{7.8.1}$$

induces an isomorphism of $IU_{\mathrm{dR}} \times \mathbb{G}_a$ with $M_{\mathrm{dR}} \subset {}_1\Pi_0$ defined in 4.2.

Proof. If u lies in $IU_{\mathrm{dR}} \subset V_{\mathrm{dR}} = \Pi \subset \mathbb{Q}\langle\langle e_0, e_1\rangle\rangle^*$, it can be written $u = 1 + \sum e_w w$, where the sum runs only over monomials of degree ≥ 3. This follows from the fact that $\mathrm{Lie}^{\mathrm{gr}}\, U_{\mathrm{dR}}$ is zero in degrees < 3. The coefficient of $e_0 e_1$ in $u \circ \rho(\lambda)$ therefore coincides with that in $\rho(\lambda)$. The coefficient of $e_0 e_1$ in dch is $-\pi^2/6 = (2\pi i)^2/24$. By (7.7.3), the coefficient in γ is therefore $1/24$, and the coefficient in $\rho(\lambda)$ is $\lambda/24$. Starting from $x = u \circ \rho(\lambda)$, one recovers λ as $24c(e_0e_1)(x)$; that x is of the form $u \circ \rho(\lambda)$ is equivalent to saying that the u defined by $x = u \circ \rho(24c(e_0e_1)(x))$ belongs to IU_{dR}. This is a closed condition, and (7.8.1) is an isomorphism with a closed subscheme of Π in which the orbit of dch is dense. This proves 7.8.

7.9

We denote by D the image of $\mathbb{G}_a$ under ρ. Thus $\rho : \mathbb{G}_a \xrightarrow{\sim} D$, and 7.8 says that the group law $\circ$ induces an isomorphism

$$\circ : IU_{\mathrm{dR}} \times D \xrightarrow{\sim} M_{\mathrm{dR}}. \tag{7.9.1}$$

This isomorphism is compatible with the gradings of affine algebras, if $O(D) = \mathbb{Q}[\lambda]$ is graded by giving λ degree -2 (codegree 2).

7.10

By 6.6 applied to N_0, the $(-1)^n c((01)^n)$, and hence their restrictions $\zeta^m(2\{n\})$ to M_{dR}, are U_{dR}-invariant. Transported from M_{dR} to $IU_{\mathrm{dR}} \times D$ by (7.9.1), the action of $u \in U_{\mathrm{dR}}$ is $(a, \lambda) \mapsto (I(u)a, \lambda)$. Viewed as functions on $IU_{\mathrm{dR}} \times D$, the $\zeta^m(2\{n\})$ are therefore inverse images of functions on D. Up to a scalar, there is only one such function in each even degree. To determine which one is $\zeta^m(2\{n\})$, it is enough to test at dch. Put $\pi_m^2 := 6\zeta^m(2)$, a definition suggested by the identity $\zeta(2) = \pi^2/6$, and $w_m^{2n} := (\pi_m^2)^n$. By 6.12, one has:

Lemma 7.11.

(i) For every n,

$$\zeta^m(2\{n\}) = \pi_m^{2n}/(2n+1)!.$$

(ii) IU_{dR} is the subscheme of M_{dR} defined by $\pi_m^2 = 0$.

Proposition 7.12. For every integer $n \geq 1$:

(i) $\zeta^m(2n)$ is a rational multiple of π_m^{2n} (the multiple is seen by evaluating at dch);

(ii) as a function on $IU_{\mathrm{dR}} \times D$, $\zeta^m(2n+1)$ is the inverse image of a function on IU_{dR}, and this function is a non-zero homomorphism from IU_{dR} to $\mathbb{G}_a$;

(iii) on abelianized groups, the $\zeta^m(2n+1)$ for $n \geq 1$ induce an isomorphism

$$U_{\mathrm{dR}}^{\mathrm{ab}} \xrightarrow{\sim} IU_{\mathrm{dR}}^{\mathrm{ab}} \xrightarrow{\sim} (\text{product of } \mathbb{G}_a).$$

Sketch of proof. The depth filtration of $\operatorname{Lie}\Pi$ is the filtration by the subalgebras "degree in $e_1 \geq i$". Let $D^i\Pi$ be the corresponding subgroups. If a subgroup Q of Π, viewed as a subgroup of ${}_0\Pi_0$, is stable under V_{dR}, it follows from (7.2.4) that it is also a subgroup of $(\Pi, \circ)$. This applies to the $D^i\Pi$.

The subgroup $D^1\Pi$ is defined by the equation $c(0) = 0$. It contains the derived group Π', defined by $c(0) = c(1) = 0$, and hence contains both IU_{dR} and dch. The quotient $D^1\Pi/D^2\Pi$ is also the quotient $D^1\Pi/D^2\Pi$ for the group law $\circ$. It is abelian, and the $c(0^{\{n-1\}}1)$ define an isomorphism of $D^1\Pi/D^2\Pi$ with a product of copies of $\mathbb{G}_a$. Because $IU_{\mathrm{dR}} \subset (\Pi, \circ)$ has a Lie algebra generated in odd degree, if n is even, $c(0^{\{n-1\}}1)$ vanishes on IU_{dR}; $\zeta^m(n)$, transported by (7.9.1) to a function on $IU_{\mathrm{dR}} \times D$, is the inverse image of a function on D, and (i) follows by the arguments of 7.10.

By (7.6.2), the $c(0^{\{n-1\}}1)$ vanish on D for n odd. Since $\zeta(n) \neq 0$, for odd $n \geq 3$ they do not vanish on IU_{dR}. This proves (ii). Assertion (iii) follows from (ii) and from the fact that the graded Lie algebra of U_{dR} has one generator in each odd degree ≥ 3.

Corollary 7.13. For all integers $a, b \geq 0$, there exist unique rational numbers $\alpha_{a,b,r}$, $1 \leq r \leq a+b+1$, such that

$$\zeta^m(2^{\{b\}}3\,2^{\{a\}}) = \sum_{r=1}^{a+b+1} \alpha_{a,b,r}\, \zeta^m(2r+1)\zeta^m(2^{\{a+b+1-r\}}). \tag{7.13.1}$$

Proof. Regard, via (7.9.1), $\zeta^m(2^{\{b\}}3\,2^{\{a\}})$ as a function on $IU_{\mathrm{dR}} \times D$. Expand it in powers of a coordinate on D:

$$\zeta^m(2^{\{b\}}3\,2^{\{a\}}) = \sum u_{2r+1}\pi_m^{2(a+b+1-r)},$$

where u_{2r+1} has codegree $2r+1$ on IU_{dR}. By 6.6 and 7.9, a left translate of u_{2r+1} differs from u_{2r+1} only by addition of a constant. It is therefore a homomorphism from IU_{dR} to the additive group. Being of weight $2r+1$, it is by 7.12 a rational multiple of $\zeta^m(2r+1)$, and 7.13 follows from 7.11.

Deus ex Machina 7.14 (D. Zagier [6]). Put

$$A^r_{a,b} = \binom{2r}{2a+2}, \qquad B^r_{a,b} = (1-2^{-2r})\binom{2r}{2b+1}.$$

Then

$$\zeta(2^{\{b\}}3\,2^{\{a\}}) = 2\sum(-1)^r(A^r_{a,b} - B^r_{a,b})\zeta(2r+1)\zeta(2^{\{a+b+1-r\}}). \tag{7.14.1}$$

This theorem suggests that the $\alpha_{a,b,r}$ of (7.13.1) are the $2(-1)^r(A^r_{a,b} - B^r_{a,b})$. How to prove this from 7.14 will be explained in §8.

7.15

The affine algebra $O(U_{\mathrm{dR}})$ of U_{dR} is dual, in the graded sense, to the enveloping algebra of $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$. This enveloping algebra is a free associative algebra on generators whose degrees are the odd integers ≥ 3. Hence the Poincaré series of $O(U_{\mathrm{dR}})$, graded by codegree, is

$$P(O(U_{\mathrm{dR}}), t) = \sum_{a\geq 0}\left(\sum_{n\geq 1} t^{2n+1}\right)^a = \left(1 - \sum_{n\geq 1} t^{2n+1}\right)^{-1} = (1-t^2)/(1-t^2-t^3).$$

If the affine algebra of the additive group $\mathbb{G}_a$, with coordinate λ, is graded by assigning codegree 2 to λ, then $P(O(\mathbb{G}_a), t) = (1-t^2)^{-1}$ and

$$P(O(U_{\mathrm{dR}} \times \mathbb{G}_a), t) = (1-t^2-t^3)^{-1}.$$

The affine algebra $O(IU_{\mathrm{dR}})$ is a graded subalgebra of $O(U_{\mathrm{dR}})$. By 7.9, one therefore has:

Proposition 7.16. The Poincaré series of $O(M_{\mathrm{dR}})$, graded by codegree, satisfies

$$P(O(M_{\mathrm{dR}}), t) \leq (1-t^2-t^3)^{-1} \tag{7.16.1}$$

(coefficientwise), with equality if and only if the morphism $I : U_{\mathrm{dR}} \to IU_{\mathrm{dR}}$ is an isomorphism, i.e. if U_{dR} acts faithfully on $\pi_1(X; 1, 0)_{\mathrm{dR}}$.

Comparing with 6.10, one obtains:

Theorem 7.17.

(i) U_{dR} acts faithfully on ${}_1\Pi_0$.

(ii) The morphism of $\mathbb{Q}$-vector spaces $\widetilde{H}(2,3) \to O(M_{\mathrm{dR}})$ is an isomorphism.

The affine algebra of $\pi_1(X; 1, 0)_{\mathrm{mot}}$ contains a $\mathbb{Q}(-1)$ as a subquotient. It follows from (i) that the action of G_{dR} on the affine algebra of $\pi_1(X; 1, 0)_{\mathrm{dR}}$ is faithful. Tannakian category theory then gives, as a consequence of (i), the following corollary.

Corollary 7.18. The affine algebra $O(\pi_1(X; 1, 0)_{\mathrm{mot}})$, which is an Ind-object of $MT(\mathbb{Z})$, generates the Tannakian category $MT(\mathbb{Z})$. More precisely, every mixed Tate motive with good reduction over $\mathrm{Spec}(\mathbb{Z})$ is obtained from subobjects in $MT(\mathbb{Z})$ of $O(\pi_1(X; 1, 0)_{\mathrm{mot}})$ by the operations of tensor products, direct sums, duals, and subquotients.

Assertion (ii) of 7.17 is equivalent to saying that the $\zeta^m(s)$ with $s_i \in \{2, 3\}$ form a basis of $O(M_{\mathrm{dR}})$.

8. Sketches of proofs

8.1

The group scheme $V_{\mathrm{dR}} = (\Pi, \circ)$ acts on ${}_1\Pi_0$ (7.1), hence on its affine algebra $O({}_1\Pi_0)$. This action is the action of $(\Pi, \circ)$ on $O(\Pi)$ by left translations (7.2.3): x sends f to $y \mapsto f(x^{-1} \circ y)$. These actions are given by (7.2.1) (7.2.2). Differentiating these formulas gives the action of $\mathrm{Lie}(V_{\mathrm{dR}}) = (\mathrm{Lie}\,\Pi, \{,\})$ on $O({}_1\Pi_0)$ and, dually, the coaction of the Lie coalgebra. This coalgebra is m/m^2, where m is the ideal of functions vanishing at the identity element 1, and the coaction is the projection onto $m/m^2 \otimes O(\Pi)$ of $f \mapsto f(x^{-1} \circ y) - f(y)$ in $m \otimes O(\Pi) \subset O(\Pi \times \Pi)$.

Goncharov gave a convenient formula for this coaction (see [5], theorem 1.2). For a word w in the alphabet $\{0, 1\}$, consider the subwords cId of $1w0$, with I non-empty: one considers all decompositions $w = w'Iw''$ of w, with I non-empty, and c, d are the letters adjacent to I in $1w0$. For each decomposition, put $w \setminus I := w'w''$. The formula for the coaction is

$$c(w) \longmapsto -\sum p_{cd}(I) \otimes c(w \setminus I), \tag{8.1.1}$$

where $p_{cd}(I)$ is the projection in m/m^2 of the following element of $m \subset O(\Pi)$:

$$p_{cd}(I) = \text{projection of } \begin{cases} 0, & c = d, \\ c(I), & c = 1,\ d = 0, \\ \text{the antipode } c(I)^* \text{ of } c(I), & c = 0,\ d = 1, \end{cases} \tag{8.1.2}$$

where, if $I = a_1 \cdots a_k$, the antipode is $(-1)^k c(a_k \cdots a_1)$.

8.2

The action of U_{dR} on $O({}_1\Pi_0)$ factors through that of $IU_{\mathrm{dR}} \subset (\Pi, \circ)$, and to obtain the coaction of the Lie coalgebra it remains to compose (8.1.1) with

$$\mathrm{coLie}(\Pi, \circ) \longrightarrow \mathrm{coLie}(IU_{\mathrm{dR}}) \longrightarrow \mathrm{coLie}(U_{\mathrm{dR}}).$$

The $c(I)$ and $c(I)^*$ of (8.1.2) matter only through their restriction to M_{dR}, taken modulo π_m^2 and the square of the maximal ideal at 1.

We choose the generators ν_{2r+1} of the Lie algebra $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$ so that, still denoting by ν_{2r+1} the image of ν_{2r+1} in the tangent space of $IU_{\mathrm{dR}} \subset \Pi$ at 1, one has

$$\langle \nu_{2r+1}, d\zeta^m(2r + 1)\rangle = 1$$

(possible by 7.12(ii)). This determines the image of ν_{2r+1} in $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}}^{\mathrm{ab}})$. If $r = a + b + 1$ and $\alpha_{a,b}$ is the coefficient of $\zeta^m(2r + 1)$ in (7.13.1), then

$$\langle \nu_{2r+1}, d\zeta^m(2^{\{b\}} 3\, 2^{\{a\}})\rangle = \alpha_{a,b}.$$

Proposition 8.3. Formula (7.14.1) remains true with ζ replaced by ζ^m.

Proof. Proceed by induction on $a + b$. If $a + b = 0$, the identity $(7.14.1)^m$ to prove reduces to $\zeta^m(3) = \zeta^m(3)$. Suppose $(7.14.1)^m$ true for $a + b < N$. This ensures that, for $a + b < N$ and $r = a + b + 1$, with the notations of 7.14, one has

$$\zeta^m(2^{\{b\}} 3\, 2^{\{a\}}) = 2(-1)^r (A^r_{a,b} - B^r_{a,b})\zeta^m(2r + 1) \quad \text{on } IU_{\mathrm{dR}}. \tag{8.3.1}$$

For $a + b = N$, if the $\alpha_{a,b,r}$ are defined as in 7.13, then, for the action of ν_{2r+1},

$$\nu_{2r+1}\zeta^m(2^{\{b\}} 3\, 2^{\{a\}}) = -\alpha_{a,b,r}\zeta^m(2^{\{a+b+1-r\}}) : \tag{8.3.2}$$

apply 7.12. For $r \le N$, compute the left-hand side by (8.1.1) (8.1.2). One has to sum over the subwords I of $w = (01)^b(001)(01)^a$, of length $2r + 1$, and surrounded by $0, 1$ or $1, 0$ in $1w0$. The $c(w \setminus I)$, restricted to M_{dR}, are $\pm\zeta^m(2^{\{a+b+1-r\}})$. The $c(I)$ or $c(I)^*$ appearing in (8.1.2), restricted to M_{dR}, are $\pm\zeta^m(2^{\{b'\}} 3\, 2^{\{a'\}})$ to which (8.3.1) applies by the induction hypothesis, or a $c((01)^m 0)$ which, on Π', is expressed by (3.5.2) as a sum of $\zeta(2^{\{b'\}} 3\, 2^{\{a'\}})$ to which (8.3.1) applies. This permits the calculation of the $\alpha_{a,b,r}$ for $r \le N$, and one finds that they have the expected value. This leaves unknown in (7.13.1) only the coefficient $\alpha_{a,b,a+b+1}$. It is determined by evaluating at dch and applying (7.14.1).

8.4

Armed with (8.3.1), one can now calculate by (8.1.1) (8.1.2) the morphisms (6.7.1). To prove 6.8, the idea is that, in the matrix obtained for (6.7.2), the dominant terms, in the 2-adic sense, come from the 2^{-2r} appearing in $B^r_{a,b}$ (7.14). If, at the target and at the source of (6.7.2), one orders the basis vectors in suitable orders corresponding under the bijection defined in 6.7, and if at the target one multiplies the basis vectors by suitable powers of 2, the matrix of (6.7.2) is a square matrix with integer coefficients whose reduction modulo 2 is triangular with 1's on the diagonal. This ensures that it is invertible.

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REFERENCES

[1] F. Brown, *Mixed Tate motives over Z*, Annals of Math., to appear in vol. 175.

[2] P. Deligne, A. B. Goncharov, *Groupes fondamentaux motiviques de Tate mixte*, Ann. Scient. Éc. Norm. Sup. **38** (2005), 1–56.

[3] L. Euler, *De summis serierum reciprocarum*, Comm. Acad. Sci. Petropol. **7** (1734/5), p.123–134, in Opera Omnia Ser. I, vol. 14, p. 73–86; see also, in the same volume of the Opera Omnia, memoir 61, p. 138–155, for an irreproachable proof, and memoir 130, p. 407–462, for an approximation of $\zeta(3)$ (p. 440).

[4] L. Euler, *Meditationes circa singulare serierum genus*, Novi Comm. Acad. Sci. Petropol. **20** (1775), p.140–186, in Opera Omnia Ser. I, vol. 15, Teubner, Berlin (1927), p. 217–267.

[5] A. B. Goncharov, *Galois symmetries of fundamental groupoids and non commutative geometry*, Duke Math. J. (2005), 1–56.

[6] D. Zagier, *Evaluation of the multiple zeta values* $\zeta(2,\ldots,2,3,2,\ldots,2)$, Annals of Math., to appear in vol. 175.

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